

PROJECT : Diabetes Dataset

```
#importing libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load dataset
data = pd.read_csv("/content/diabetes.csv")
```

```
#display first five rows
print("display 5 rows:")
print(data.head())
```

```
display 5 rows:
  Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI   \
0            6     148             72           35          0  33.6
1            1      85             66           29          0  26.6
2            8     183             64            0          0  23.3
3            1      89             66           23          94  28.1
4            0     137             40           35         168  43.1

  DiabetesPedigreeFunction  Age  Outcome
0                0.627     50         1
1                0.351     31         0
2                0.672     32         1
3                0.167     21         0
4                2.288     33         1
```

```
#Basic inforamation
```

```
print("Basic information:")
print(data.info())
```

```
Basic information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Pregnancies           768 non-null   int64
 1   Glucose               768 non-null   int64
 2   BloodPressure         768 non-null   int64
 3   SkinThickness         768 non-null   int64
 4   Insulin               768 non-null   int64
 5   BMI                   768 non-null   float64
 6   DiabetesPedigreeFunction 768 non-null   float64
 7   Age                   768 non-null   int64
 8   Outcome               768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

```
#check for missing values
```

```
print("\nMissing Values:")
print(data.isnull().sum())
```

```
Missing Values:
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction 0
Age               0
Outcome           0
dtype: int64
```

```
#Drop duplication
```

```
data = data.drop_duplicates()
```

```
#Basic Statistics
print("\nStatistics Summary:")
print(data.describe())
```

```
Statistics Summary:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	846.000000

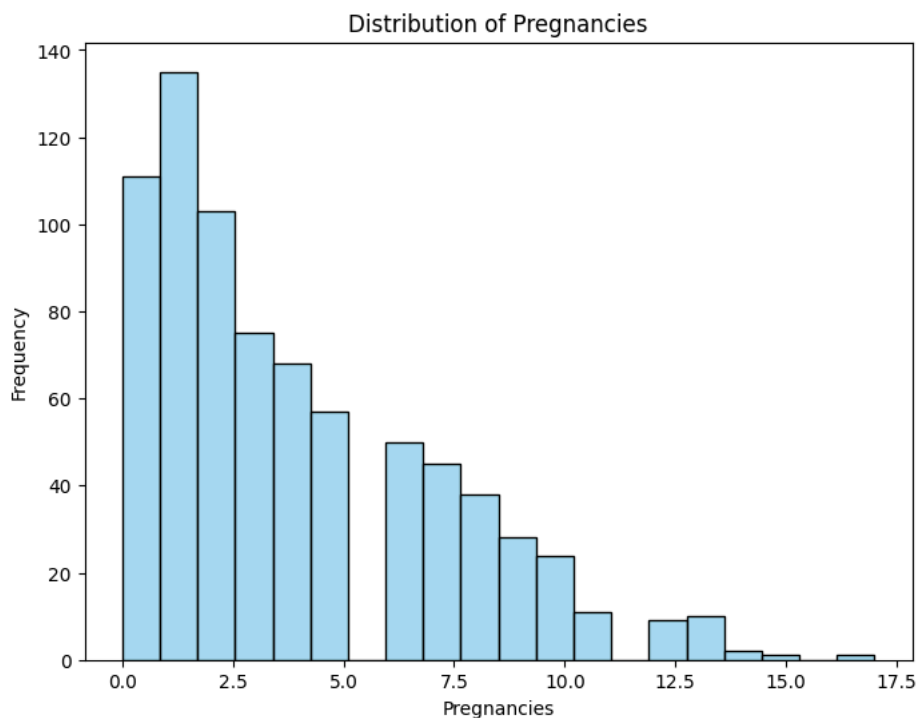
	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

```
#Data cleaning
```

```
data.columns = data.columns.str.strip()
data = data.dropna()
```

```
#correlation for numeric feature
corr = data.corr(numeric_only=True)
```

```
#data visualization
#distribution of sleep duration
plt.figure(figsize=(8, 6))
sns.histplot(data['Pregnancies'], bins=20,color = 'skyblue')
plt.title('Distribution of Pregnancies')
plt.xlabel('Pregnancies')
plt.ylabel('Frequency')
plt.show()
```

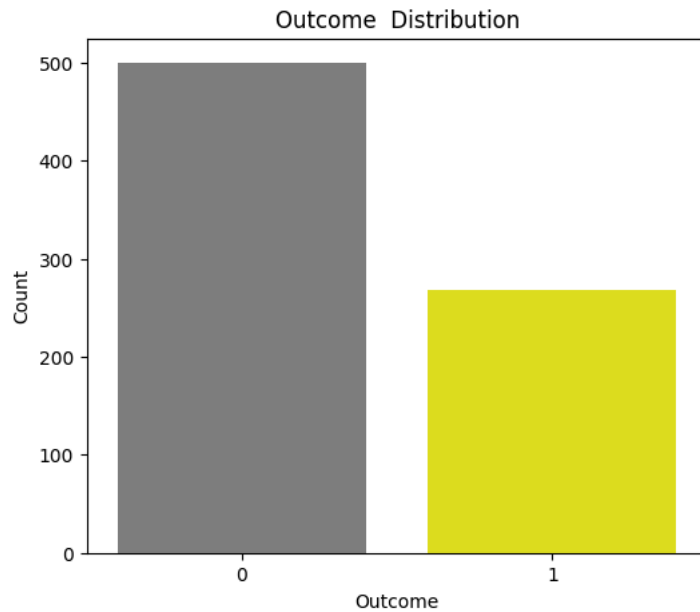


```
#Bar chart for Outcome
outcome_counts = data['Outcome'].value_counts()
plt.figure(figsize=(6,5))
sns.barplot(x=outcome_counts.index, y=outcome_counts.values, palette=['grey', 'yellow'])
plt.title('Outcome Distribution')
plt.xlabel('Outcome')
plt.ylabel('Count')
plt.show()
```

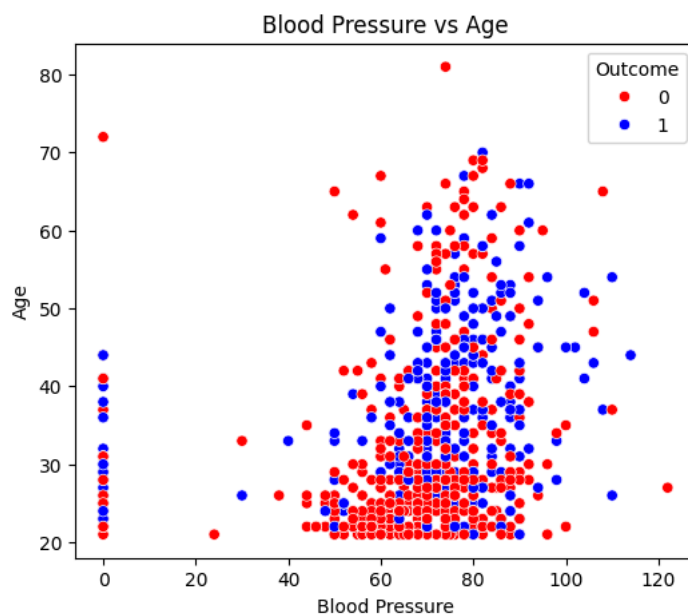
```
/tmp/ipython-input-2691532282.py:4: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

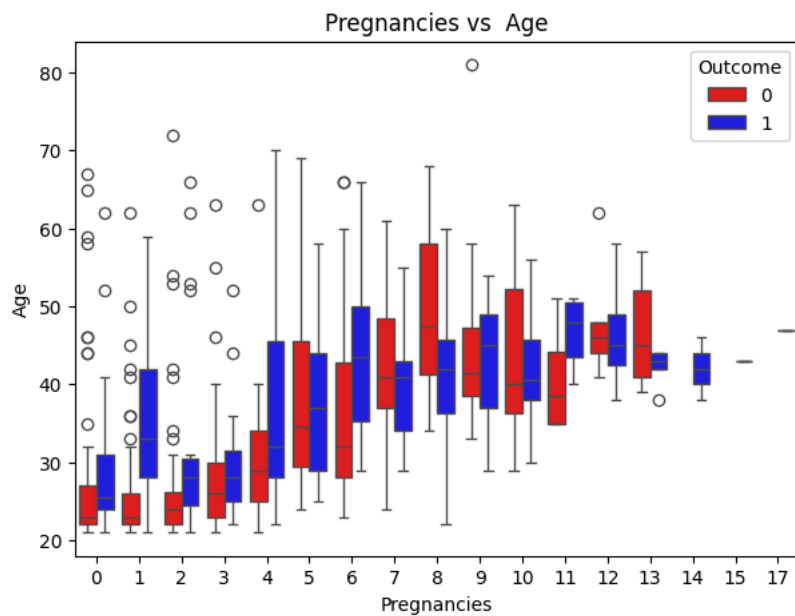
```
sns.barplot(x=outcome_counts.index, y=outcome_counts.values, palette=['grey', 'yellow'])
```



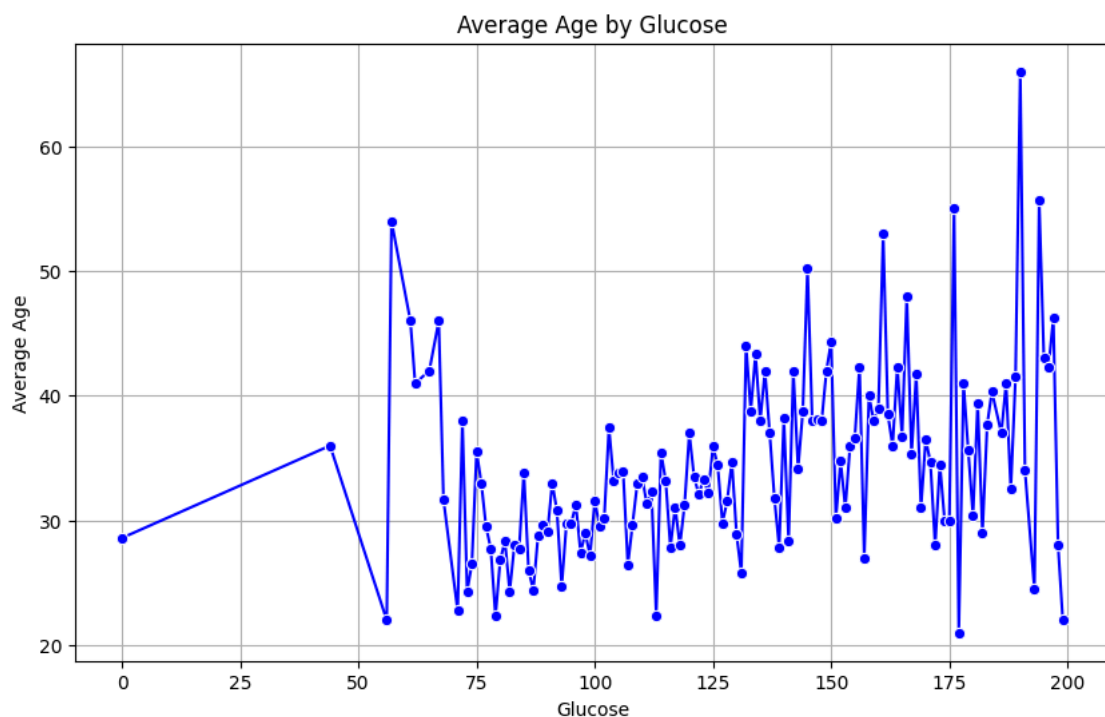
```
#scatter plot:bloodpressure vs age
plt.figure(figsize=(6,5))
sns.scatterplot(x='BloodPressure', y='Age', data=data, hue='Outcome', palette=['red', 'blue'])
plt.title('Blood Pressure vs Age')
plt.xlabel('Blood Pressure')
plt.ylabel('Age')
plt.show()
```



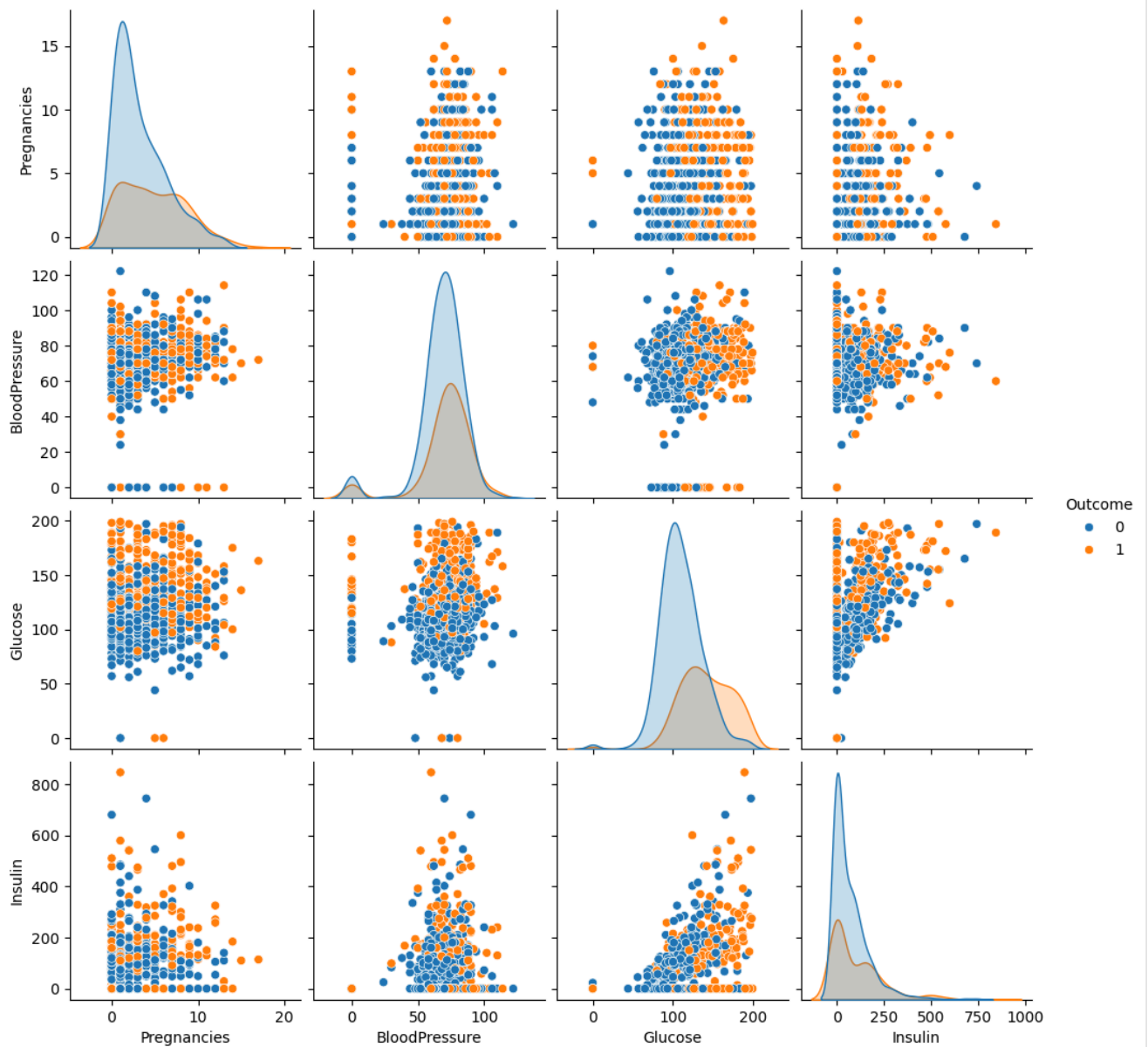
```
#box plot pregnancies and Age
plt.figure(figsize=(7,5))
sns.boxplot(x='Pregnancies', y='Age', hue='Outcome', data=data, palette=['red', 'blue'])
plt.title('Pregnancies vs Age')
plt.xlabel('Pregnancies')
plt.ylabel('Age')
plt.show()
```



```
#line plot : age va bmi
data = pd.read_csv("/content/diabetes.csv") # Reload the dataset
avg_Age = data.groupby('Glucose')['Age'].mean().reset_index()
plt.figure(figsize=(10,6))
sns.lineplot(x='Glucose', y='Age', data=avg_Age, marker = 'o', linestyle = '-', color = "blue" )
plt.title('Average Age by Glucose')
plt.xlabel('Glucose')
plt.ylabel('Average Age')
plt.grid(True)
plt.show()
```



```
#pairplot
sns.pairplot(data, vars=['Pregnancies',
                        'BloodPressure',
                        'Glucose',
                        'Insulin'],
             hue = 'Outcome')
plt.show()
```



```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
#dataset
data = pd.read_csv("/content/diabetes.csv")
#compute corelation matrix
corr = data.corr(numeric_only= True)
#plot heat map
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, fmt = ".2f", cmap='coolwarm')
plt.title('Correlation Heatmap = Diabetes dataset')
plt.show()
```